

Exploring Transfer Learning and Convolutional Networks in The High-Throughput Analysis of Medicinal Plant Morphology

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ABSTRACT

Automated image-based medicinal plant research utilizes deep learning methods to provide innovative classification methods together with evaluation procedures for medicinal properties. The identification technique delivers superior results than traditional plant recognition because it depends on specialized expertise and human verification yet produces lengthy processes along with incorrect interpretations. This report demonstrates an automated photo analysis approach for medicinal plant classification through CNN execution. The method used recognizes color, texture, and form, among other basic leaf features, thanks to extensive training of deep learning models.. This enables precise species classification. Transfer learning technologies based on pre-trained networks support accurate outcomes in addition to needing smaller datasets for network development. transfer learning improves the model's accuracy while minimizing the demand for intensive data collection. This method has the potential to significantly improve plant research by facilitating the quicker and more precise identification of therapeutic plants in field research as well as the herbal medicine industry. Additionally, by providing a non-invasive way to track plant species and their therapeutic qualities, the system contributes to biodiversity conservation. All things considered, this study establishes the foundation for incorporating AI into botanical research, enabling the effective and scalable examination of plants for medicinal purposes

Keywords: *MICC, CASIA, and UCID, TPR, FPR, Image Processing, and Python.*

1. INTRODUCTION

Medical identification of plant species continues to be a challenge in Picture handling and PC Vision people group fundamentally due to their extensive existence, intricate structure, and unpredictability various classes related to nature. Due to these regular intricacies, it is exceptionally bothersome to perform typical division or element extraction or on the other hand consolidating shape, surface and variety highlights which brings about moderate accuracy with reference datasets. Despite the fact that there are some methods joining worldwide and neighbourhood highlight descriptors arrives at state of the workmanship exactness in characterizing clinical s, still there is a requirement for a hearty and proficient framework to consequently distinguish and perceive medical species on a larger scale in an environment that is complex. Saith and Kane proposed a method for recognizing medical images in which two needed, one from the leaf and one from the medical field. This method calls for to identify it, the user must place a black cloth behind the medical device. These are not practical and is badly arranged for the client to involve this technique in genuine time situation.

2. RELATED WORK

The implementation of software programs depends heavily on literary analysis during development processes. Before starting development of the device it is essential to understand important time factors and cost reduction potential and commercial application stability requirements. The operating systems and programming languages need identification before moving forward with growth procedures when all conditions are satisfied. The device development process requires programmers to seek multiple kinds of external assistance after starting their programming work. Advanced programmers together with other resources found on books and websites offer assistance for software program development. The system design process begins by expanding the proposed tool following analysis of the mentioned problems. The evaluation of improvement requirements holds prime importance within the goal development branch. The application development process requires literature evaluation as its core essential component for every project execution. Organizations must determine and analyze time elements, aid necessities, human sources, economics, and organizational talents before developing equipment alongside their associated designs. Research into meeting all necessary factors leads to detailed specifications for software program programming along with working system and software transfer selections. Development of equipment and related features begins first followed by development of their features Deep convolutional neural networks processed 1.2 million high-resolution pictures in ImageNet LSVRC-2010 challenge until they sorted them into their thousand unique classes. The evaluation information demonstrated our model reaching 37.5% and 17.0% top-1 and top-5 misclassification rates that outpaced all previous leading achievements. The brain organization with its 60 million boundaries and 650,000 neurons includes five convolutional layers combined with max-pooling layers as well as three completely connected layers that terminates with a 1000-way SoftMax layer. To speed up the learning process we employed non-saturating neurons together with an optimized GPU implementation of convolution calculations. A variation of this model won the ILSVRC-2012 competition by achieving 15.3% error on the top five tests better than the 26.2% accomplished continuously best passage. [1]. The network achieves current state-of-the-art results with its Inception concept highlighted as the main feature of the architecture during the ImageNet Enormous Scope Visual Challenge for Recognition 2014 (ILSVRC14). A methodically designed approach allowed us to increase both depth and width of the network without exceeding our predetermined computational capabilities. The network architecture was designed for maximizing quality. For the ILSVRC14 challenge we submitted Google Net as our entry. This implementation of the One Our submission for ILSVRC14 uses a particular incarnation known as Google Net, a 22 layers profound organization, the nature of which is evaluated in the context of detection and classification. [2].

Contemporary speech recognition systems handle temporal variations in speech with hidden Markov models while Gaussian mixture models (GMMs) determine boundary fit of individual phones or short coefficient windows that represent voice input. Neural networks process frame coefficients as inputs and produce back probabilities over Gee states as output for measuring the fit in a different manner[3-5]. The existing framework uses Procedural identification and recognition methods for medicinal plant species within forest areas. Input of images enables identification and classification of medical plant species as per this research work[6][7]. The appearance of crop leaves establishes the basis for their particular diseases. The system uses high resolution digital photography for picture acquisition followed by k-means clustering for separating damaged leaf sections after preliminary treatment. The various machine learning algorithms are then applied to these, and ordered in light of their variety and surface highlights[8]. A correlation in light of exactness between different AI algorithms include Naive Bayes, K-Nearest Neighbours, and Multinomial Strategic Relapse to accomplish most extreme precision[9]. Utilizing traditional methods to identify medicinal plant strategies or present-day procedures like picture handling, can have a few issues and Intricacy of Species Distinguishing proof: Natural plants frequently have a place with various species with unobtrusive contrasts apparently. This intricacy can make exact recognizable proof testing, particularly for non-specialists or in places with a lot of biodiversity[10].

Latest discourse acknowledgment frameworks utilize stowed away Markov models (Gee) to manage the fleeting inconstancy of discourse and Gaussian blend models (GMMs) to decide how well each condition of each well fits an edge or a short window of casings of coefficients that symbolizes the sound input. To assess the fit in a different way, make use of a feed-forward neural network that processes a number of frames coefficients as information and produces back probabilities over Gee states as results Profound brain organizations (DNNs) that have many stowed away It has been demonstrated that layers and new methods for training can outflank GMMs on an assortment of discourse acknowledgment benchmarks, once in a while overwhelmingly. This article gives an outline of this progress and addresses the common perspectives on four exploration bunches that have recently utilized DNNs for acoustic modelling with success in recognition of speech [3]. Latest discourse acknowledgment frameworks utilize stowed away Markov models (Gee) to manage the fleeting inconstancy of discourse and Gaussian blend models (GMMs) to decide how well each condition of

each well fits an edge or a short window of casings of coefficients that symbolizes the sound input. To assess the fit in a different way, make use of a feed-forward neural network that processes a number of frames coefficients as information and produces back probabilities over Gee states as results Profound brain organizations (DNNs) that have many stowed away It has been demonstrated that layers and new methods for training can outflank GMMs on an assortment of discourse acknowledgment benchmarks, once in a while overwhelmingly. This article gives an outline of this progress and addresses

the common perspectives on four exploration bunches that have recently utilized DNNs for acoustic modelling with success in recognition of speech [14].

Latest discourse acknowledgment frameworks utilize stowed away Markov models (Gee) to manage the fleeting inconstancy of discourse and Gaussian blend models (GMMs) to decide how well each condition of each well fits an edge or a short window of casings of coefficients that symbolizes the sound input[15-18]. To assess the fit in a different way, make use of a feed-forward neural network that processes a number of frames coefficients as information and produces back probabilities over Gee states as results. Deep brain organizations (DNNs) that have many stowed away It has been demonstrated that layers and new methods for training can outflank GMMs on an assortment of discourse acknowledgment benchmarks, once in a while overwhelmingly. This article gives an outline of this progress and addresses the common perspectives on four exploration bunches that have recently utilized DNNs for acoustic modelling with success in recognition of speech [19].

Existing framework model has done Programmed recognizable proof and acknowledgment of therapeutic plant species in conditions like timberlands. Image processing is used to identify and categorize sun medical in this paper[20]. Diseases of crops based on how their leaves look as shown in Table 1. The pictures are taken through a high goal computerized camera and after pre-handling, are used k-means clustering to extract the diseased leaf portion[21]. The various machine learning algorithms are then applied to these, and ordered in light of their variety and surface highlights. A correlation in light of exactness between different AI algorithms include Naive Bayes, K-Nearest Neighbours, and Multinomial Strategic Relapse to accomplish most extreme precision. Utilizing traditional methods to identify medicinal plant strategies or present-day procedures like picture handling, can have a few issues and Intricacy of Species Distinguishing proof: Natural plants frequently have a place with various species with unobtrusive contrasts apparently[22][23]. This intricacy can make exact recognizable proof testing, particularly for non-specialists or in places with a lot of biodiversity.

Table 1: Table of Collected Accuracy data

Author(s)	Year	Proposed Work	Proposed Algorithm	Accuracy
Govindharaj, Kapil Rajput, Navin Garg, Vinay Kukreja, Rishabh Sharma	2024	CNN-RF hybrid model for disease detection in rice crops (rust, Hispa, mildew)	CNN (for feature extraction), Random Forest (RF for classification)	Not specified
Zubair Saeed, Ali Raza, Ans H. Qureshi, Muhammad Haroon Yousaf	2021	Generalized approach to detect diseases in multiple crops using CNN	Convolutional Neural Network (CNN)	Not specified
Amandeep Kaur, Kalpna Guleria, Naresh Kumar Trivedi	2021	Review of rice leaf disease detection methods		Not specified
S. Ramesh, V. Mohanavel, S. U. Anitha G Diwakaran, Maheswaran	2022	Detection of rice diseases using GLCM texture feature algorithm	Gray Level Co-occurrence Matrix (GLCM)	Not specified
Vinay Kukreja, Rishabh Sharma, Rishika Yadav	2023	Hybrid CNN- SVM model for detecting rice sheath rot disease	Feature extraction using CNN	Not specified
Murala Praveena et.al.	2023	Analyzing DL and ML methods for rice leaf disease detection	SVM, CNN	Outperforms state-of-the-art methods on Eye PACS and Messidor datasets
Naresh Cherukuri, G. Ravi	2021	HKNN and CNN for early	K-Nearest Neighbors	Kappa score: 0.90

Kumar, Ongole Gandhi, Venkata Sai Krishna Thotakura, Dama NagaMani CMAK Zeelan Basha		detection of disease in rice.	(KNN), CNN)	(APTOS), 0.88 (Eye PACS)
Syed Taha et al.	2022	Using SRGAN for data augmentation and DenseNet 169, VGG16 for rice disease classification	SRGAN DenseNet121. DenseNet169, MobileNetV2, VGG16	Not specified
Sarder Tanvir Ahmed, Shomtirtha Barua, Md. Fahim-Ul-Islam, Amitabha	2024	Optimized BEIT model for early rice leaf disease detection	Optimized BEIT (CNN+ Transformer), GRAD-CAM	Not explicitly stated in the provided abstract.

3. MATERIALS AND METHODS

CNN is a strong man-made reasoning device in design grouping. CNN characterizing clinical picture classes. The CNN architecture is planned with four convolutional layers. layer with various sifting window sizes is viewed as which works on the speed furthermore, exactness in acknowledgment. A stochastic pooling strategy is executed which joins the upsides of both max and mean pooling strategies[24-26]. Preparing is acted in various clusters to know the strength of gigantic preparation modes expected for SVM's. In Cluster V of preparing, the preparation is performed with four sets of information and amplifying the grouping rate. This CNN architecture has better training and validation accuracies than the different models[27]. A less measure of preparing and approval misfortune is seen with the proposed CNN architecture. The goal of using medical image processing techniques home grown plants can incorporate a few significant objectives: Plant Recognizable proof and Arrangement: Picture handling can help with the programmed ID and grouping of therapeutic plants based on their visual highlights. Botanists will particularly benefit from this, herbalists and researchers who require precise plant identification in either in the wild or herbal markets. Convolutional Brain Organizations (CNNs) are amazing assets for picture recognition, categorization, and other spatial data-related tasks Nonetheless, similar to any innovation, CNNs have specific impediments and limitations. CNNs have a few drawbacks, CNNs can be computationally serious, particularly while managing huge scope datasets or profound architectures. CNN training necessitates significant computational assets, including elite execution GPUs, which can be exorbitant Picture procurement can be characterized as the demonstration of getting a picture from sources. This should be possible by equipment framework like cameras and datasets and furthermore some encoder's sensors additionally happen in this cycle[28].

The primary objective of image pre-processing is enhancement of information like picture that lessens the reluctant twists or enhances some features, we can simply say that the unwelcome interference with the image[29][30]. It is a piece of the decrease cycle in correspondingly in which a starting set of raw data is broken down into more sensible gatherings. A pixel is transformed into a labelled image through this process from the picture. This can process the important through this procedure. Fragments not a whole picture. The errand of distinguishing what precisely in the picture. That process will occur by the model is prepared to comprehend the various classes. For egg: you may prepare a model to perceive the three distinct creatures in the picture.

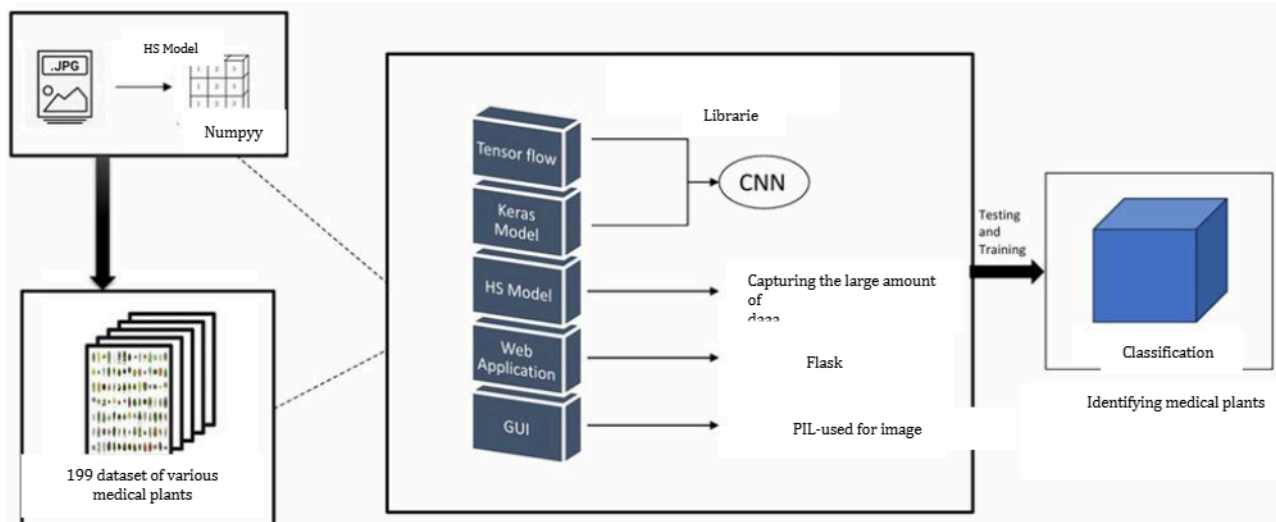


Figure 1: System Architecture

The Figure 1 system architecture makes use of a dataset of 199 images of plant leaves in JPG format. These images are converted into HS (hyperspectral) model data and NumPy arrays for processing. TensorFlow and Keras are used to build and train a Convolutional Neural Network (CNN). HS Model helps in capturing a large amount of data. Flask is used to develop a web application for deploying the model. PIL (Python Imaging Library) is used for image processing in the GUI. The dataset is trained and tested through the CNN model. The trained model does classification to identify different medicinal plants. The output is a classification of the medicinal plant from the input image of a leaf. The output is represented visually in a cuboid, possibly representing the output layer of the network, concealing the output layers. The training and testing phase plays a crucial role, ensuring that the model has accuracy and generalizes well to unseen data.

4. PROPOSED METHODOLOGY

The work collects medicinal plant images (betel, curry, tulsi, mint, neem, and Indian beech) from the Kaggle dataset prior to image size unification. Apply data augmentation (rotation, flipping, brightness variation, etc.) to increase dataset diversity. And the second thought was deep learning model selection & Training which use the VGG-16 deep learning model due to its lightweight nature and efficiency in mobile applications. It Split dataset into training (70%), validation (15%), and testing (15%) sets and train the model using transfer learning techniques for faster convergence. And third thought is evaluation & optimization helps to evaluate the model using accuracy, precision, recall, and F1-score. It applies fine-tune hyperparameters to optimize performance and achieve a final accuracy of 98.3%.

A. Data Collection & Preprocessing:

The Data Collection & Preprocessing Module obtains images from the Kaggle dataset followed by resizing and data augmentation steps until it saves the prepared images for training purposes. The initial process entails compiling a diverse collection of medicinal plants images that come from both botanical gardens and herbarium collections and open-source databases of plant images. The gathered dataset needs to contain pictures of multiple plant sections including leaves and flowers alongside stems to demonstrate unique elements. The model receives added benefits from data augmentation procedures which duplicate plant images through transformations such as rotations and flips and scale adjustments with color modification to create diverse visual examples of plants presented across different lighting scenarios and camera perspectives and backgrounds. Human beings manually assign plant and medicinal labels to images through their expertise or by working with botanists to accomplish the task. The process of normalizing both images for standard ranges prepares them for deep learning compatibility through resolution standardization. Additional processing techniques including cropping and brightness alterations help expand the diversity of the collected data. The labels receive numerical encoding through one-hot encoding methods that are frequently used for classification objectives. The dataset divides into training, validation and test components to ensure the model trains through separate data before evaluation which boosts its prediction accuracy for medicinal plants.

B. Model Selection, Evaluation and Optimization:

The deep learning system performs automatic computer vision methods for medicinal plant analysis from images. Model selection begins the process which needs appropriate selection of deep learning architecture for image classification using Convolutional Neural Networks (CNNs) because of their proven effectiveness in visual data processing and classification

abilities. The approach of transfer learning lets researchers enhance performance using restricted datasets by applying pre-trained models VGG16, ResNet and Inception to optimize their plant dataset through prior task knowledge acquisition. The evaluation process begins after selecting the model and applying training procedures. An independent test set should be used for model evaluation because it determines how well the model performs on new unfamiliar images. A combination of accuracy, precision, recall, F1-score with the confusion matrix enables researchers to assess both the model's plant species classification effectiveness and its occurrence of misclassification errors. Model evaluation allows researchers to pinpoint weak points that affect its performance by determining if it produces erroneous positive or negative predictions.

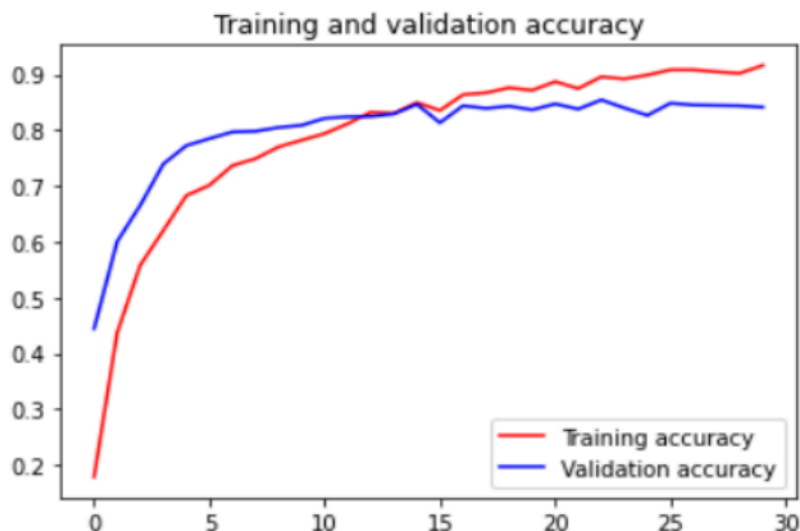


Figure 2: Training and Validation Accuracy of Plant Analysis

The target of model optimization consists of adapting the model structure to generate superior performance outcomes as shown in Figure 2 and 3. The adjustment of learning rate and batch size along with network layer count functions as part of optimization strategy.

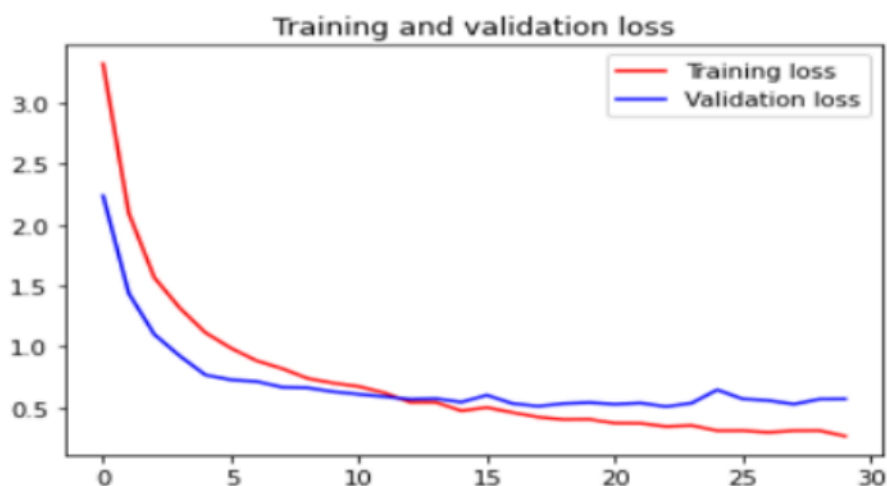


Figure 3: Training and Validation loss of Plant Analysis

The dropout and L2 regularization serve as essential techniques for preventing overfitting so that the model shows satisfactory performance throughout training and testing phases. The optimization process aims to enhance model classification accuracy for medicinal plants by minimizing errors which leads to effective processing of genuine plant image variations.

C. Medicinal Plant Analysis:

Science-based methods with image-based analysis and deep learning power plant identification and medicinal property evaluation and classification processes. The process of deep learning uses trained models to examine plant images which then identifies species and evaluates medicinal value by analyzing visual traits of the plants. First among medicinal plant

analysis steps is clear identification of species. Deep learning models consisting mostly of Convolutional Neural Networks recognize essential plant features through images to classify plants by species. The proper species identification in herbal medicine needs precise plant recognition because wrong species use would lead to incorrect therapeutic applications. In addition to species identification, medicinal plant analysis can also extend to quality and maturity assessment. By analyzing images of plants at different stages of growth, the model can assess the maturity of the plant, which is important for determining its medicinal efficacy. For example, certain medicinal compounds in plants might be more potent at specific stages of growth, so knowing the optimal time for harvesting is essential. Furthermore, medicinal property prediction can be integrated into the analysis process. Some deep learning models can be designed to correlate visual features of the plant with known medicinal properties, allowing for predictions about the plant's potential therapeutic uses. This can aid researchers in identifying new medicinal plants or confirming the efficacy of traditional medicinal practices. Combining botanical expertise with cutting-edge image recognition methods via medicinal plant analysis results in automated plant assessment, identification, and classification. This is a vital research tool for botanical studies, herbal medicine, and biodiversity preservation proposes.

D. Deployment and Real-Time Analysis:

Deep learning models require deployment with practical user-friendly systems through which healthcare professionals can utilize them for practical tasks. After concluding the optimization process the trained model can provide its functions to researchers as well as field botanists and members of the general public. During deployment the trained model gets merged into mobile applications web-based platforms and software systems which allow real-time user interactions with the model. The trained model is incorporated by web-based platforms along with mobile applications for providing instant fashion model interaction which enables effective user decision-making in remote locations with scarce expert botanist resources. Real-time analysis happens through the model processing plant images automatically which results in immediate feedback. Through this process a user can input an image of a medicinal plant which triggers a deep learning model to give plant identification results along with quality evaluation and possible medicinal usage information. Real-time processing through the system provides essential functionality for various fields such as herbal medicine and biodiversity monitoring and conservation because it improves decision-making speed.

At the same time the deployment may incorporate an edge computing system that executes the model on devices such as smartphones or drones independent of server dependency. The system yields faster processing results particularly when operating under restricted internet conditions hence supporting field research and conservation activities. Real-time deployment makes deep learning practical for medicinal plant identification thus enabling users to utilize artificial intelligence for better plant species analysis including medicinal properties and harvesting time decisions. The connection between technology enables investigators to perform quick accurate taxonomic species determination without invasive methods for crime investigations and industrial monitoring and conservation efforts.

E. Classification Model

This produces the probabilities of each class as an output. The third module is Model Evaluation & Optimization Module helps in testing the model with unseen data and Computes accuracy, precision, recall & F1-score with Tune hyperparameters if needed.

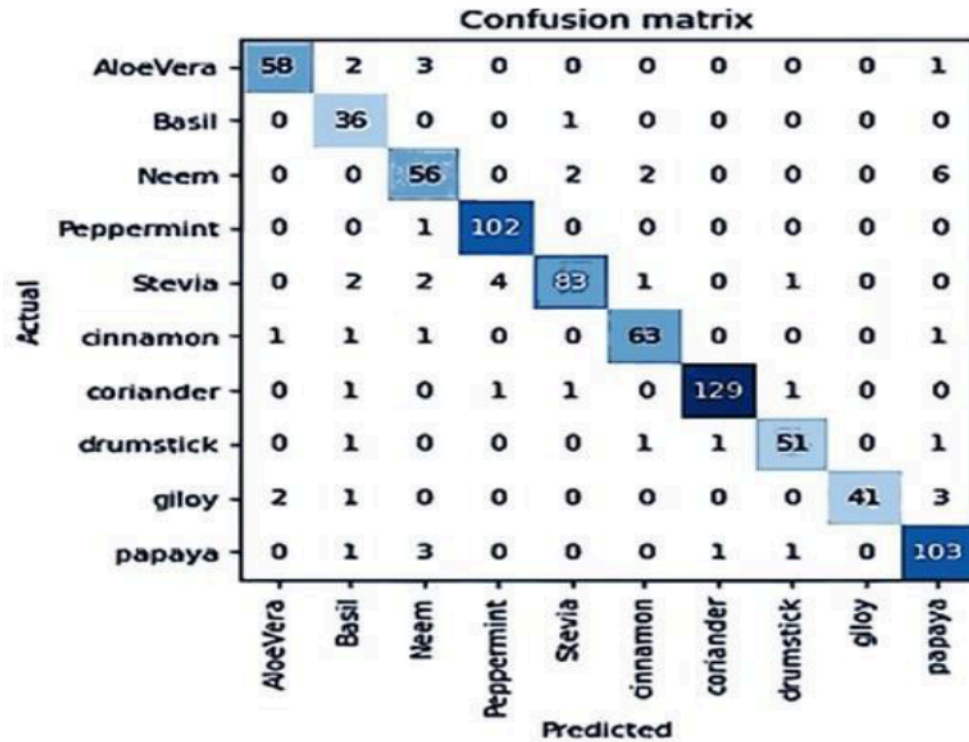


Figure 4 Confusion matrix 1

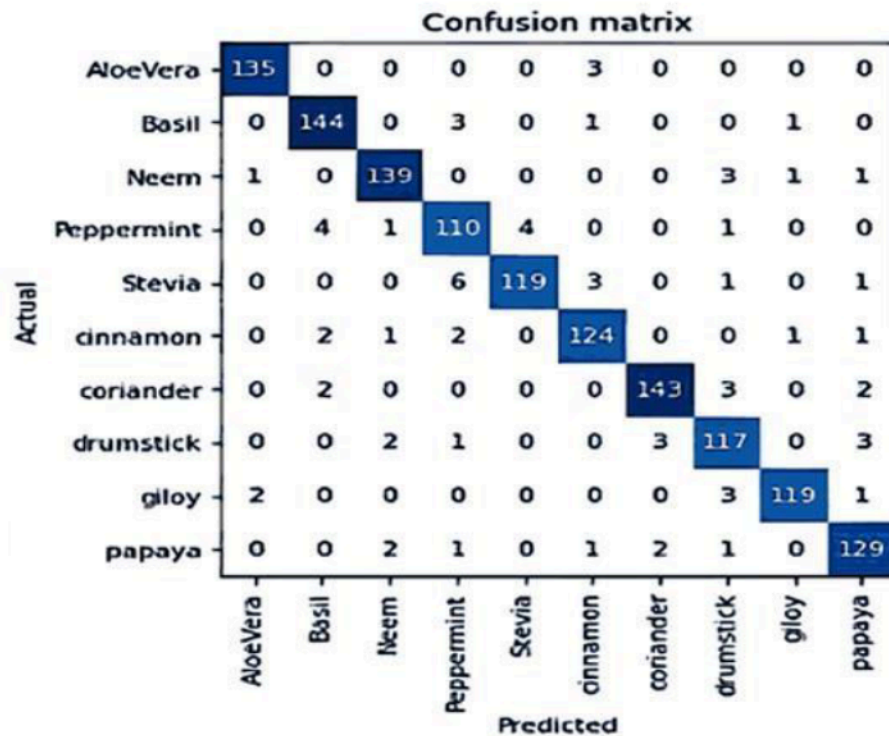


Figure 5 Confusion matrix 2

The above Figure 4 and Figure 5 matrix represents the performance of a model classifying medicinal plant images of Aloe Vera, Basil, Neem, Peppermint, Stevia, Cinnamon, Coriander, Drumstick, Giloy, and Papaya. It the actual plant category .And it predicted plant category. The diagonal elements (from top-left to bottom-right) show the correct classifications. The off- diagonal elements show incorrect classifications.

E. Classification Model

This produces the probabilities of each class as an output. The third module is Model Evaluation & Optimization Module helps in testing the model with unseen data and Computes accuracy, precision, recall & F1-score with Tune hyperparameters if needed.

Table 2: Classification model Validation

MODEL	TESTING	TRAINING	PERFORMANCE	VALIDATION
VGG	0.7973	0.7987	0.9884	0.9884
RESNET	0.9995	1	1	1
MOBILE NET	0.9995	0.9989	1	1
CNN	1	0.9958	1	1

The above Table 2 is all about testing, training, validation, and the performance of different DL models first and third one will come under the vgg and resnet-based model, while another two will be mobile net and cnn part.

5. RESULT & DISCUSSION

The main objective of this research is to establish real-time identification methods for medicinal plants. The necessary photos of medicinal leaves for this challenge were obtained from Kaggle. All photographs received inclusion in a specific size range because they initially had varying dimensions. Geometrical augmentation was implemented on leaf images because the collected data lacked sufficient training capacity for the DL model. The DL model receives training through 15 epochs from augmented input data. The main objective of this research work involves achieving real-time plant medication recognition as shown in Table 3. The needed images for medicinal leaves were obtained from Kaggle to complete this challenge. The obtained photographs underwent sizing treatment due to their variable dimensions. The obtained data collection fell short for model training thus geometrical image augmentation became necessary. After using augmented data during training the DL model achieved 15 epochs and delivered a test accuracy of 98.33 and training precision of 97.36 and training precision of 99.32% as illustrated in Figs. 1 and 2. The research reveals that the developed TL system successfully detects medicinal plants based on its overall average score surpassing 97%. Industry progress in medicinal plant identification and classification has excelled through development of image processing and deep learning approaches for identification and classification. Medical plants requiring precise automatic identification procedures because they serve as natural treatment options while posing potential risks through inaccurate identification. Ongoing research and implementation of these technologies will support worldwide enhancement of patient results and sustainable healthcare practice implementation.

Considering the vital role that medicinal plants play in offering natural cures for a range of ailments, as well as the possible hazards that come with misidentification, it is imperative that precise and automated procedures be developed. These technologies support biodiversity preservation and traditional knowledge preservation in addition to improving the accuracy and efficiency of plant identification. Globally enhancing patient outcomes and promoting sustainable healthcare practices will be greatly aided by ongoing study and application.

Table.3 Some types of medical plants datasets and its uses

Scientific Name	Local Name	Jpg	Part used	Medical use
Aloe barbadensis miller	Aloe Vera		Leaf	Wound Healing

Ocimum basilicum	Basil		Leaf	Inflammatio n
Azadirachta indica	Neem		Leaf	Fever
Mentha x piperita	Peppermin t		Leaf	Headache
Stevia rebaudiana	Stevia		Leaf	Anti-diabetic
Lurus cinnamomu m	Cinnamon		Stem	Anti-cancer
Coriandrum sativum	Coriander		Seeds	Indigestion
Moringa oleifera	Drum stick		Flower s	Control Blood- pressure
Tinospora cordifolia	Giloy		Leaf	Anti- dandruff
Carica papaya	Papaya		Papay a sap	Antiseptic

In Figure 6, Display the name of the plant species which is given as the input image. This can be done with in the given datasets only. From Figure 6 the output is displayed and the uses of that specific species is given.

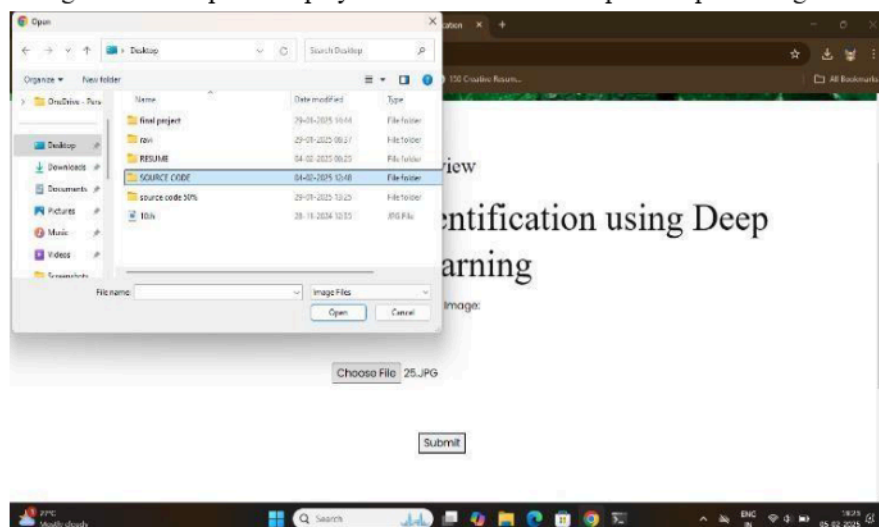


Figure 6: Selecting image from the data that was provided in source code.

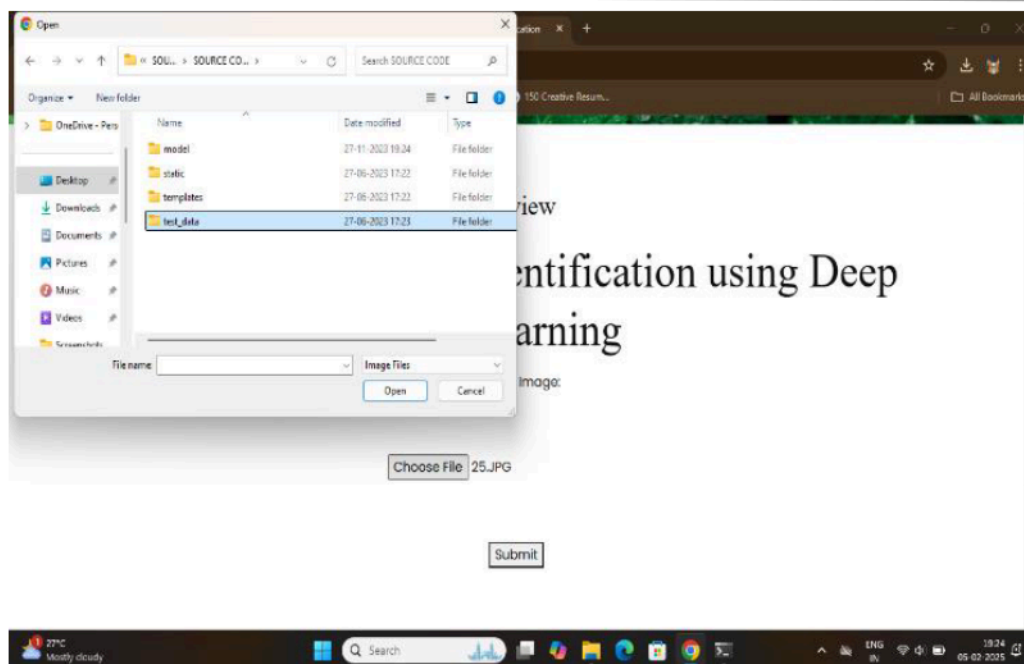


Figure.7: Selecting the test_data from the data sets given.

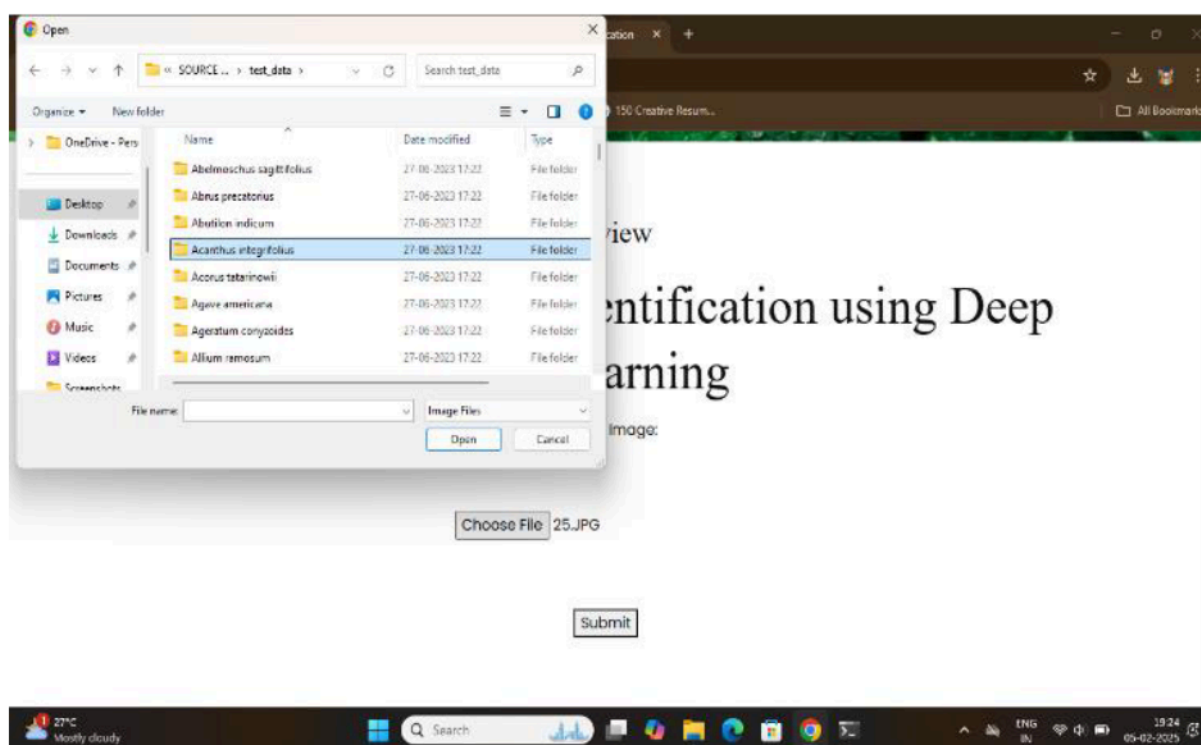


Figure.8: Selecting the plant data set that was provided.

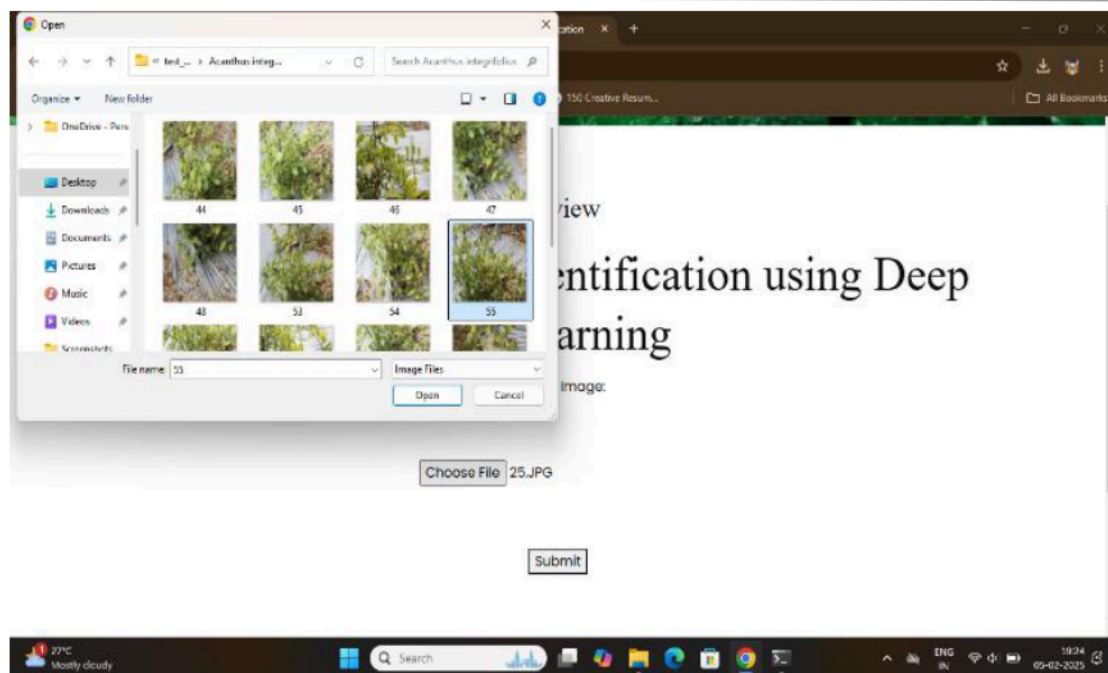


Figure.9: Uploading the plant images.



Figure.10: Submitting the image of the plant data.



Figure.11: Display of the plant scientific name.

Then uploading the plants image data sets (i.e Figure 7-9). Submitting the plant image as shown in Figure 10. The output displays the name of the plant species as shown in Figure 11.

6. CONCLUSION

A widely inclusive way to deal with spice distinguishing proof and quality assessment is given by the proposed framework to picture handling based therapeutic spice examination. It made the most common method of analysing spices, making it more precise and transparent while advancing constant assessment and enlightening undertakings in the field of home developed drug. An all-encompassing approach to herb identification and quality evaluation is provided by the proposed system for image processing-based medicinal herb analysis. It smoothed out the most common way of examining spices, making it more exact and open while supporting continuous examination and instructive endeavors in the field of home grown medication. The application has demonstrated significant potential in assisting both professionals and enthusiasts in the field of herbal medicine. By leveraging advanced image processing and machine learning algorithms, users can easily identify medicinal plants in real-time, which enhances the accessibility and accuracy of herbal plant identification.

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